

Heatmap

A color-based matrix showing values (often correlations or feature values)

- Hierarchical clustering helps reorder data
- Heatmap helps visualize that reordered data

Combined visualization is called a: *Clustered Heatmap*

Working

Step 1:

Hierarchical clustering is applied on:

- Rows (customers)
- Columns (features)

Step 2:

Data is rearranged so similar items are together

Step 3:

Heatmap colors show values

You see clusters visually

- Similar rows appear together
- Similar columns appear together

2. No need to predefine K

Unlike K-Means

3. Pattern discovery

- Customers with similar RFM patterns group together
 - Blocks of similar color → similar customers
 - Dendrogram → cluster hierarchy

Hierarchical clustering groups similar data points, and heatmaps help visualize these groups using colors, often along with dendrograms.

Clustering helps us reorganize the heatmap so that similar patterns are grouped together, making hidden structures visible.

What is Hierarchical Clustering?

A method of clustering where data points are grouped step-by-step into a tree-like structure called a dendrogram.

Two types

◊ Agglomerative

Start with each point as its own cluster → keep merging

◊ Divisive

Start with all points → keep splitting

Example:

A B C D E

Step-by-step merging:

(A,B) (C,D) E

→ ((A,B),(C,D)) E

→ (((A,B),(C,D)),E)

This builds a tree (hierarchy)

What is a Dendrogram?

A tree diagram showing how clusters are formed.

- Each merge = a branch
- Height of branch = distance at which merge happened

How distance between points is calculated?

Euclidean distance

$$d(x,y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

Example

Point A = (2,3)

Point B = (5,7)

Distance:

$$\sqrt{[(5-2)^2 + (7-3)^2]}$$

$$= \sqrt{[9 + 16]}$$

$$= \sqrt{25} = 5$$

Distance between CLUSTERS

Once clusters form, we need distance between clusters. (We start calculating cluster-to-cluster distances AFTER first merge)

Linkage Methods

◊ 1. Single Linkage (nearest)

Distance = closest points between clusters

$$\text{dist}(A,B) = \min(\text{distance between any points})$$

◊ 2. Complete Linkage (farthest)

Distance = farthest points

$$\text{dist}(A,B) = \max(\text{distance between points}) - \text{Creates compact clusters}$$

◊ 3. Average Linkage

Average distance between all points

$$\text{dist}(A,B) = \text{average of all pair distances}$$

◊ 4. Ward Method (MOST USED)

Minimizes increase in variance
`linkage(data, method='ward')`

It merges clusters such that total cluster variance increases minimally
It Gives:

- Balanced clusters
- Compact groups

Example

We take 4 points :

A = 1
B = 2
C = 8
D = 10

Step 1: Compute initial distances

Distance = $|x - y|$

	A	B	C	D
A	0	1	7	9
B	1	0	6	8
C	7	6	0	2
D	9	8	2	0

Step 2: First merge

Smallest distance = 1 (A, B)

New cluster:

(A,B)

Step 3: Recompute distances (Single Linkage)

Now clusters:

(A,B), C, D

Compute distances

(A,B) $\rightarrow$ C

- A $\rightarrow$ C = 7
- B $\rightarrow$ C = 6

min = 6

(A,B) $\rightarrow$ D

- A $\rightarrow$ D = 9
- B $\rightarrow$ D = 8

min = 8

C $\rightarrow$ D

already = 2

New distance matrix

	(A,B)	C	D
(A,B)	0	6	8

C	6	0	2
D	8	2	0

Step 4: Second merge

Smallest distance = 2 (C, D)

New cluster:

(C,D)

Step 5: Recompute again

Now clusters:

(A,B), (C,D)

Compute distance

Check ALL combinations:

- $A \rightarrow C = 7$
- $A \rightarrow D = 9$
- $B \rightarrow C = 6$
- $B \rightarrow D = 8$

Single linkage = min = 6

Final matrix

	(A,B)	(C,D)
(A,B)	0	6
(C,D)	6	0

Step 6: Final merge

Merge at distance = 6

Final Dendrogram

